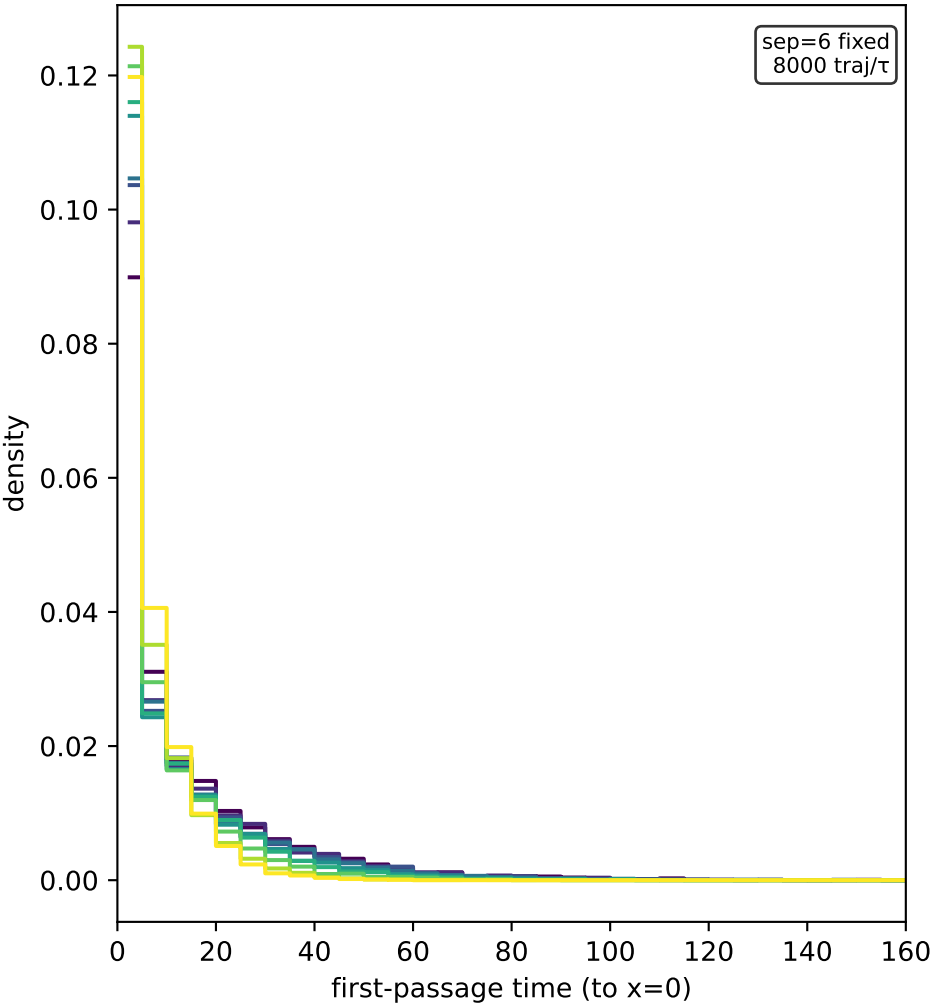


Fig 3 — Reducibility in conservative 1D is definitional (Arm A), not empirical. Honest negative: τ -dial is equilibrium in $U_{\text{eff}}(\tau)$. Frontier = v4.

Engine FPT distributions (τ family)
(the measured dial)



Reducibility is definitional here

- Arm A established: causal force is conservative
 $F_{\text{ce}} = -\nabla U_{\text{eff}}$ ($U_{\text{eff}} = -\beta T_c S_c$)
- Therefore at fixed τ the engine dynamics *are* equilibrium relaxation in $U_{\text{eff}}(\tau)$
- In 1D: every force is curl-free by geometry.
'Engine $\equiv$ readout' is true by definition, not measurement.
- Presenting duplicate-SDE KS $p \geq 0.05$ as a 'finding' would be circular. We do not do that.

The empirical anchor for thermodynamic indistinguishability is Part D (fig 5): $\langle L_z \rangle \approx 0$ for the engine (vs nonzero for the Rotor2D positive control).

The sharp question 'does the τ -knob leave a fingerprint no static potential can mimic?' receives answer NO in this conservative 1D regime.

Real frontier = v4: non-conservative force (curl $\neq 0$) in 2D+, where a non-vacuous reducibility test exists.